

OrbiterCrew and MPU - beta 0.2081026b (Orbiter 2024)

OrbiterCrew: people with physiology, a space suit and a jet pack. MPU / EMPU: an open and a sealed electric 8x8 mobile platform, with a charging station. Early beta.

Install

1. A clean **Orbiter 2024** (x86). Start it with **Orbiter_ng.exe** (the D3D9 client); the inline-graphics orbiter.exe is not supported.
2. Launchpad, tab Modules: tick **XRSound** (all sound goes through it).
3. Unpack the archive into the Orbiter folder (it only adds files).
4. Two settings are needed for the EMPU cabin:
 - `D3D9Client.cfg: NearClipPlaneMode = 1` (otherwise holes appear in the cabin walls) and `ShadowMapMode = 2` (otherwise the Sun shines through the hull and the cabin lamps are invisible).
 - `Orbiter_NG.cfg: EnableLocalLights = TRUE, MaxLights = 8` (the cabin lamps and headlights).
 - Recommended: `D3D9Client.cfg: ShaderCacheUse = 1` (the client keeps its compiled shaders: much faster starts after the first).
5. Scenarios: folder **MPU** (start with "MPU - Earth, KSC"; its description lists the controls). In any of them the person can stand up (F), put the suit on (K) and use the suit computer.

Scenarios

Scenario	What to do
MPU - Earth, KSC	Drive the EMPU, charge it at the station, swap cells, couple the open MPU
MPU - Moon, Brighton Beach	The same on regolith in vacuum
MPU - Mars, Olympus	The same under a thin red-dust sky
MPU - Earth, crash test	Speed limiter off, 130 km/h into the charging station
MPU - Earth, KSC, Delta Glider	Board a stock Delta Glider through its nose airlock, fly it, get out (see "Boarding other ships" below)

On foot

- W/S walk, A/D turn, Q/E side step, Shift run, Space jump. F1 eyes / outside view; right mouse held looks around.
- K suit on/off (off only in breathable air), L lamps, B takes / drops the jet pack.
- Look at a thing within reach: its actions show beside the aim; the mouse wheel picks one, F does it. Alt gives the cursor and marks every place of an action within 8 m with an orange square.
- Inside a cabin F1 switches between the eyes and a view from behind (it stops at the walls).
- Language (English / Russian): the suit computer's left MFD, page Options, switch RU | EN - it sets the suit computer and the interaction menu at once and is remembered (`LANGUAGE` in

`Config\OrbiterCrew\OrbiterCrew.cfg`). The EMPU's screens have their own switch (terminal, SETUP: RUS / ENG).

- In a cabin the head sways with the machine's inertia and a light road shake at speed: `SHAKE` in the same file (1 as modelled, 0.5 half, 0 none).

Platforms

- W/S drive and brake (reverse from standstill), A/D steer, Shift+W full power, Space brake, B parking brake, Caps Lock platform up/down.
- EMPU: sealed cabin with an airlock, a yoke, a touch terminal (NAV, POWER, LIFE / ENV, GEAR, MOTOR, SETUP; RUS / ENG; speed limiter) and cabin lights.
- Charging: F at the station's cable reel, then at the machine's power socket (front bumper, right). Cells come out and go in at the bays and at the station's rack.
- Crashes damage wheels and the hull; above 50 km/h they injure the people inside.

The suit computer (H, eyes view), the interaction menu and the messages are English or Russian (see Language above); the scenario texts are English.

Binaries only. The suit cannot be sat in the EMPU's seat yet.

Boarding other ships

An OrbiterCrew person can board ships that are not part of OrbiterCrew: the stock Delta Glider, and any other ship. The ship's own module is never changed.

How it works

- Walk up to the ship and look at its airlock (or docking port) within reach. Its actions show in the menu beside the aim (mouse wheel picks, F does; Alt shows the places of actions as orange squares).
- **open / close the airlock** - OrbiterCrew presses the ship's own keys for you (as if you pressed them with the ship in focus) and reads its door animations back.
- **go in** (the airlock open) - the person is aboard: the ship gets the focus and the view, and you fly it as usual. The person breathes the cabin air.
- **Getting out** - press **F** with the ship in focus, standing on the ground with the airlock open. The person steps out at the airlock and gets the focus back.

Any ship, with no file

A ship with no file of its own (see below) can be boarded at its **first docking port**: the menu shows "go in" there (no doors to work). Classes with a menu of their own are left out in

`Config\OrbiterCrew\Ships\_default.cfg`:

```
EXCLUDE OrbiterCrew\TVehicles\MPU EMPU Tantra    ; class names that start so
OFF                                                ; (on a line of its own) no default boarding
at all
```

Adding a ship: one file

Create `Config\OrbiterCrew\Ships\<ClassName>.cfg`, where `<ClassName>` is the ship's class name (the part after the colon in a scenario line, e.g. `GL-01:DeltaGlider` -> `DeltaGlider.cfg`; a class in a subfolder,

Vendor\Ship, goes to Ships\Vendor\Ship.cfg). UTF-8 text; ; starts a comment.

Line	Meaning
LABEL	The place's name in the menu: the Russian name, a vertical bar, the English one (shown with LANGUAGE en)
NODE x y z	The airlock's outer opening, ship frame, metres
REACH r	How far from the person's eyes it can be used, metres
OPEN K 17 0 11	To open: pairs of a key (letter A-Z, pressed on the ship) and the animation it moves
CLOSE 0 11 K 17	To close, the same way
IN 11 17	The animations that must be open to go in and out
EXIT x y z	Where the person steps out, ship frame (put on the ground under it)

Without OPEN the menu shows only "go in".

The animation numbers are the ship's own: the order in which its module creates its animations (CreateAnimation), from 0. They can be found in the ship's source, or from its author.

Example: the stock Delta Glider

```
LABEL носовой шлюз | nose airlock
NODE 0 -0.49 10.40
REACH 3.2
OPEN K 17 0 11 ; K the nose cone (animation 17), 0 the outer door (11)
CLOSE 0 11 K 17
IN 11 17
EXIT 0 0 12.5
```

Notes

- Out only on the ground, stopped, with the airlock open.
- A person aboard is saved with the scenario (RIDE <ship name> in the person's block) and is aboard again when it is loaded.
- Scenario to try: **MPU - Earth, KSC, Delta Glider**.